Business Question and Visualization Report

Global Olympic Sports Performance Analysis (1992–2026)

Date: 24 September 2026	**Team ID:** PNT2026TMID092418
Project Name: Global Olympic Sports Performance Analysis	**Maximum Marks:** 5 Marks

Business Questions and Visualisation Overview

Visualization development refers to the process of creating graphical representations of historical Olympic data to facilitate understanding, analysis, and strategic decision-making. Below are the 8 core business questions and Tableau visual outputs designed for NOC strategists, grant analysts, and media storytellers.

1. Business Question: Which sports categories generate the highest average medals per athlete?

Visualization Type: Horizontal Bar Chart showing Avg Medals per Athlete by Sport Category (Tableau Desktop).

Insight & Visual Summary: Rowing leads all categories at **6.916 average medals per athlete**, followed closely by Shooting at **6.731**. Demonstrates high per-athlete yield return for NOC investment.

2. Business Question: How do host nations perform during their home Olympic Games compared to other hosts?

Visualization Type: Comparative Bar Chart comparing Medal Totals Across Olympic Hosts.

Insight & Visual Summary: Japan leads home host nation performance with **105 host medals**, followed by the United States with **100 medals** and China with **93 medals**.

3. Business Question: How do overall total medal counts compare across top competing nations from 1992 to 2026?

Visualization Type: Ranked Bar Chart comparing total medal tallies by country.

Insight & Visual Summary: The United States leads globally with **735 cumulative medals**, followed by China with **696 medals**, establishing top-tier athletic benchmarks.

4. Business Question: What is the relationship between sports capital investment and actual medal efficiency?

Visualization Type: Global Sports Investment vs. Medal Performance Bubble Chart.

Insight & Visual Summary: Identifies high-efficiency, overperforming nations such as **Slovakia, Georgia, and Ethiopia**, which achieve strong medal tallies relative to modest capital budgets.

5. Business Question: Which geographic regions demonstrate the highest average medal efficiency percentage?

Visualization Type: Average Medal Efficiency Percentage by Region Pie & Donut Chart.

Insight & Visual Summary: South America leads regional efficiency at **49.64%**, followed by Europe/Asia at **45.98%**, guiding targeted international sports development grant allocations.

6. Business Question: How have total medals awarded fluctuated year-over-year from 1992 to 2026?

Visualization Type: Trend of Total Medals Awarded Over the Years Line Chart.

Insight & Visual Summary: Traces medal totals swinging between a low of **305 in 2000** to a peak of **363 in 2020**, and recording **334 in 2026**.

7. Business Question: How are medals distributed regionally between Summer and Winter Olympic Games?

Visualization Type: Regional Medal Efficiency and Distribution by Sport Donut/Stacked Chart.

Insight & Visual Summary: Europe and Asia maintain historic dominance in Winter Games, while North and South America hold strong footprint in Summer sports.

8. Business Question: How has athlete participation and medal distribution evolved across gender demographics?

Visualization Type: Dual-Axis Line & Area Chart tracking male vs. female athlete participation over time.

Insight & Visual Summary: Shows consistent historical growth in female athlete representation, reaching near 50-50 gender parity across modern Olympic cycles.